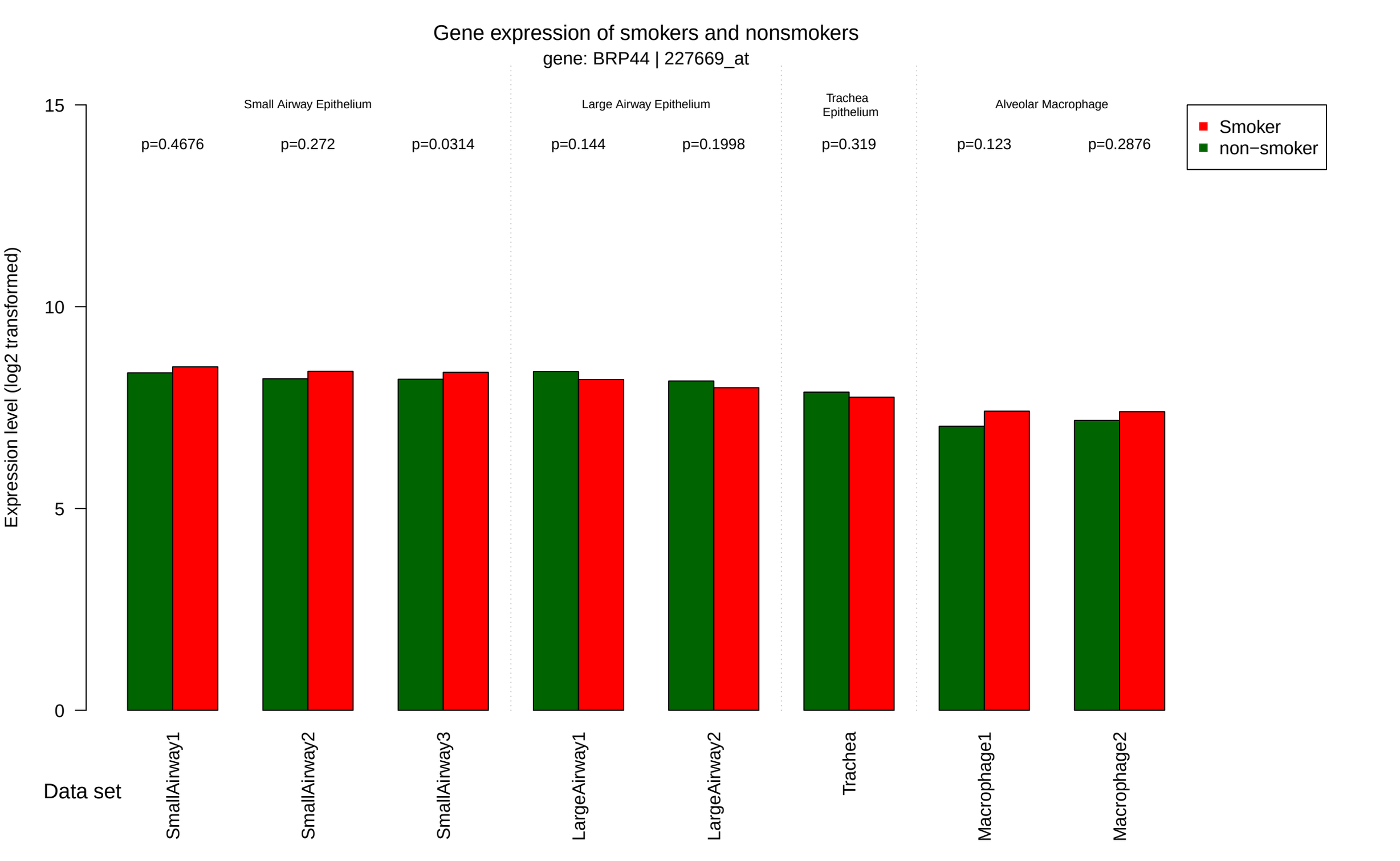
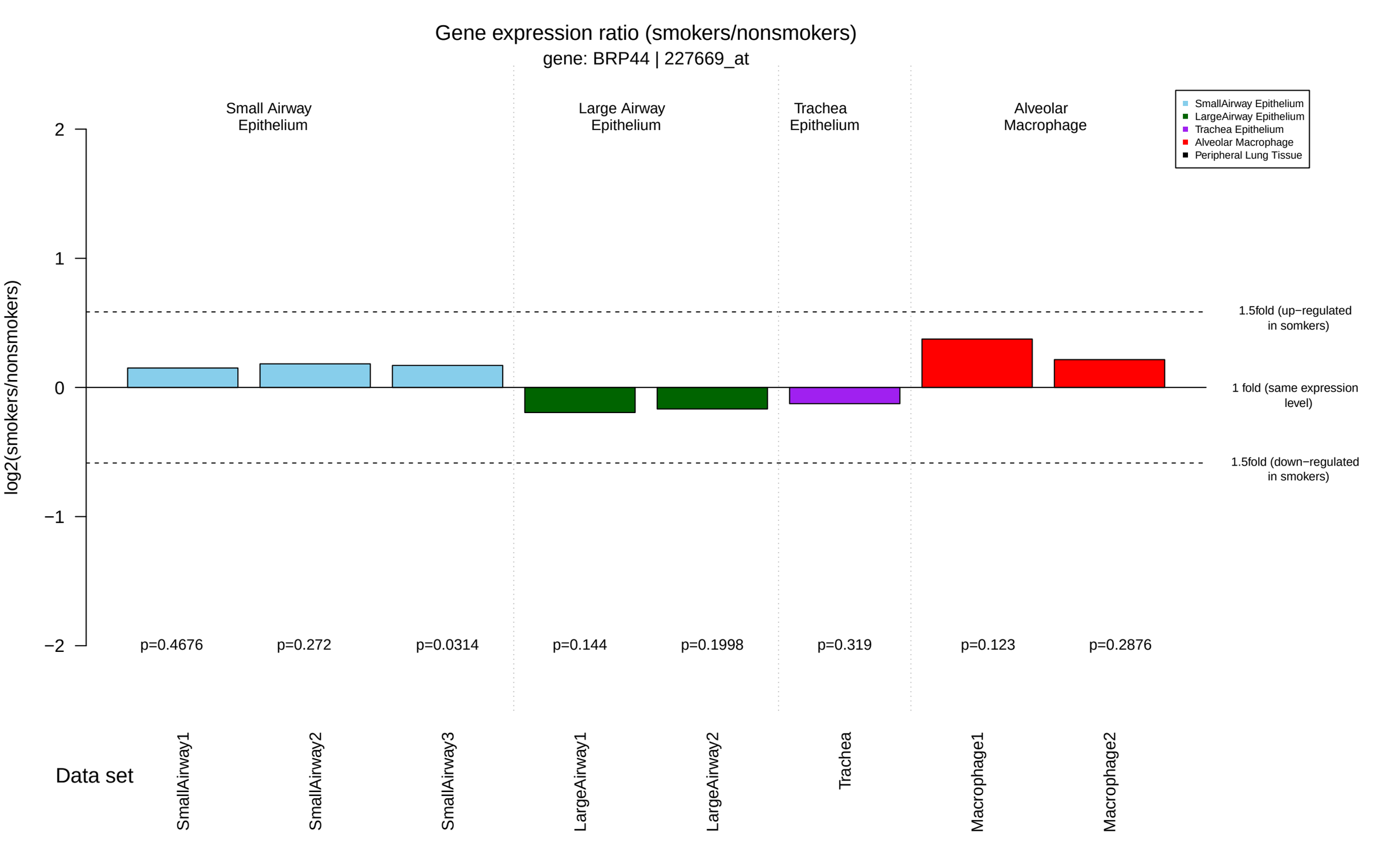
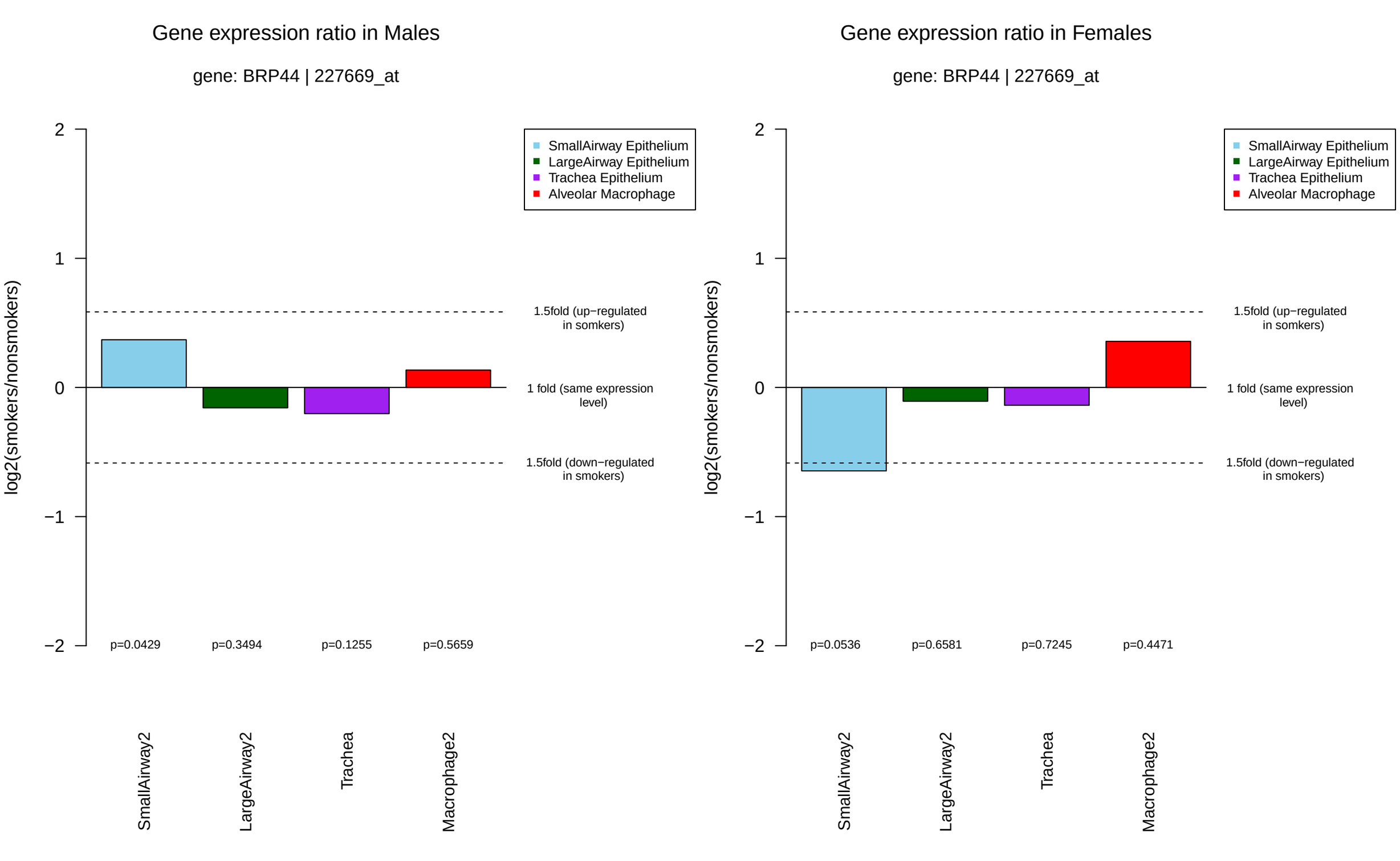
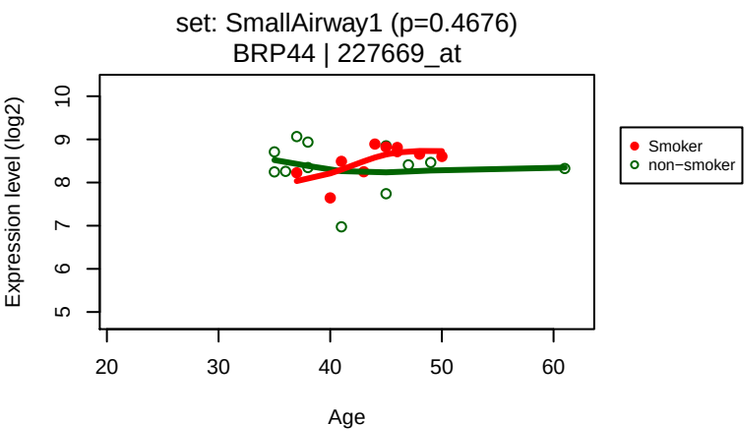
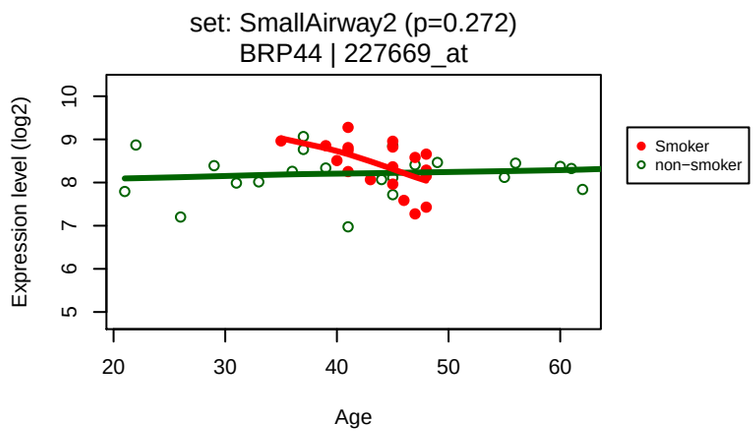
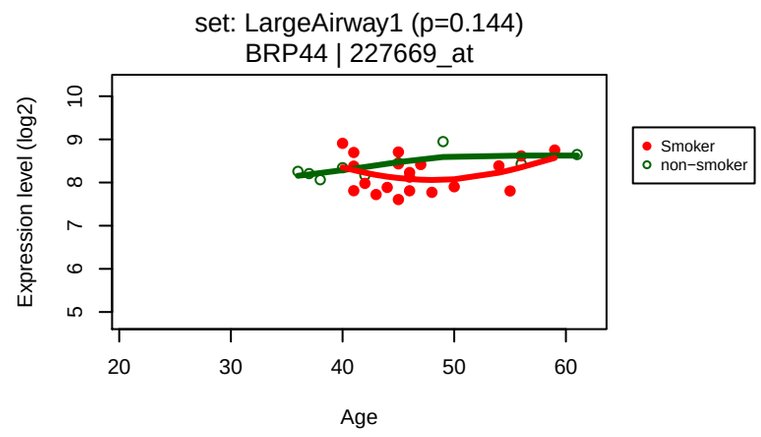

Gene expression of smokers and nonsmokers
gene: BRP44 | 227669_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4676
p=0.272
p=0.0314
p=0.144
p=0.1998
p=0.319
p=0.123
p=0.2876
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

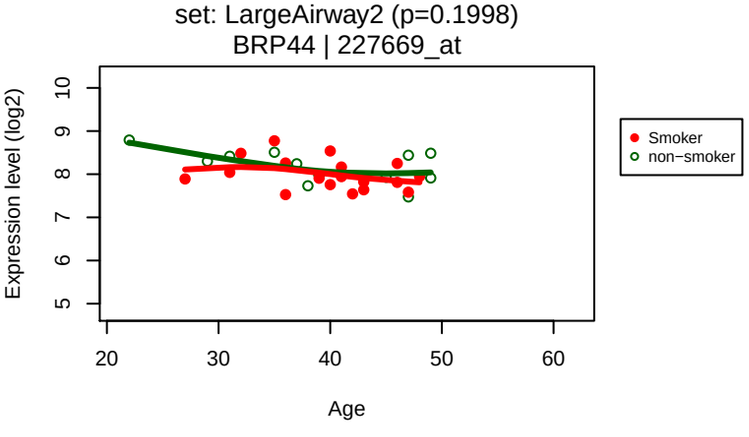

Gene expression ratio (smokers/nonsmokers)
gene: BRP44 | 227669_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4676
p=0.272
p=0.0314
p=0.144
p=0.1998
p=0.319
p=0.123
p=0.2876
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

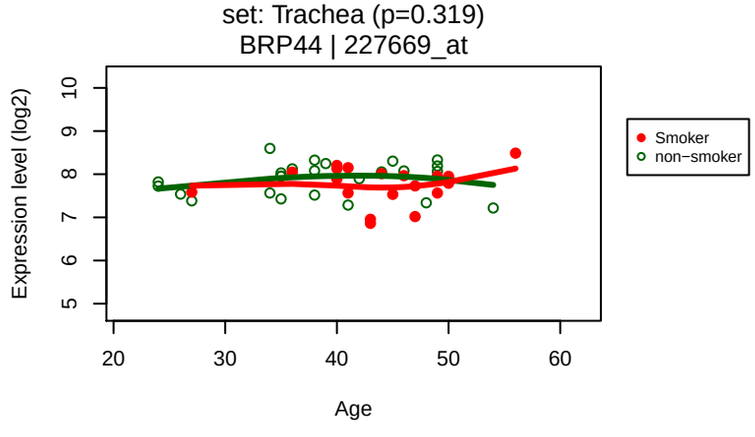

Gene expression ratio in Males
gene: BRP44 | 227669_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0429
p=0.3494
p=0.1255
p=0.5659
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BRP44 | 227669_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0536
p=0.6581
p=0.7245
p=0.4471
SmallAirway2
LargeAirway2
Trachea
Macrophage2

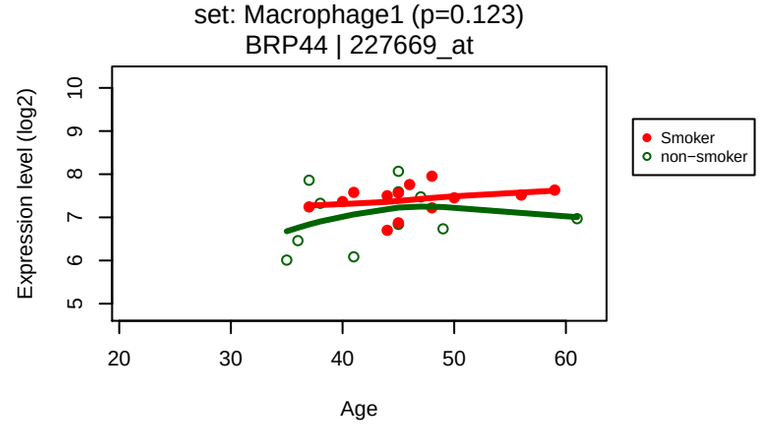

set: SmallAirway1 (p=0.4676)
BRP44 | 227669_at
Expression level (log2)
Age
Smoker
non−smoker

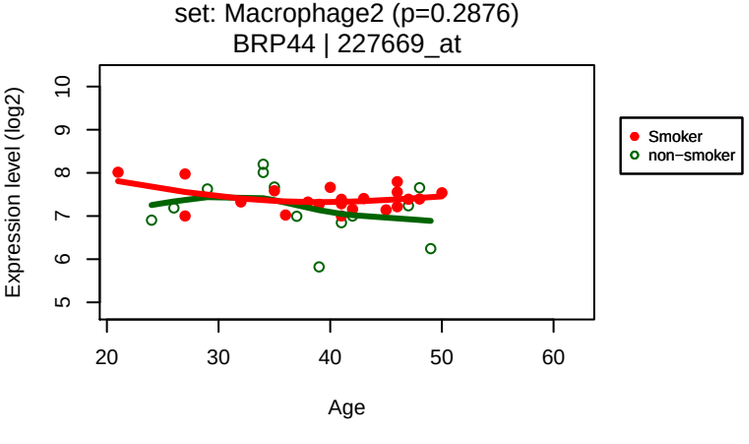

set: SmallAirway2 (p=0.272)
BRP44 | 227669_at
Expression level (log2)
Age
Smoker
non−smoker

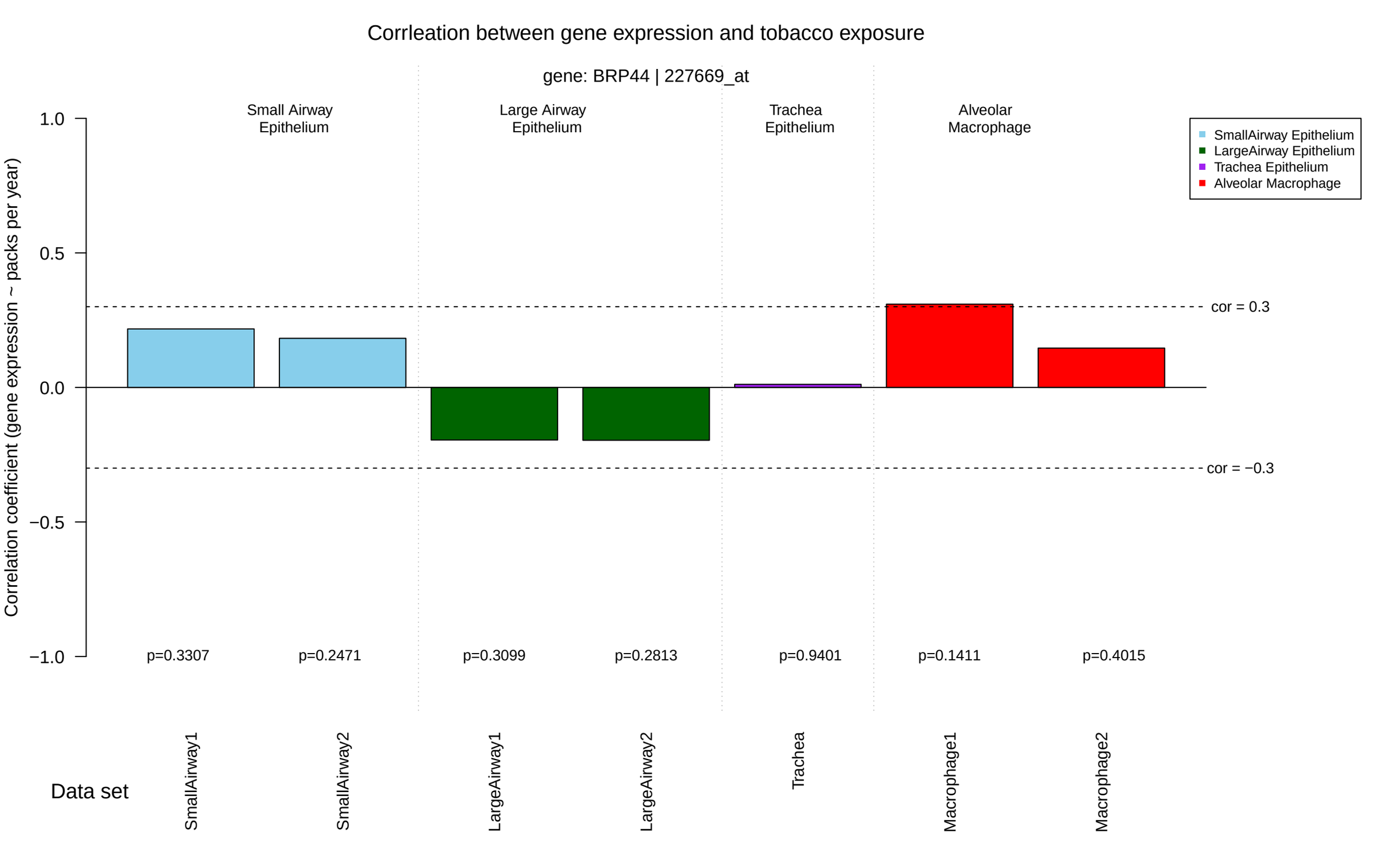

set: LargeAirway1 (p=0.144)
BRP44 | 227669_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1998)
BRP44 | 227669_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.319)
BRP44 | 227669_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.123)
BRP44 | 227669_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2876)
BRP44 | 227669_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BRP44 | 227669_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3307
p=0.2471
p=0.3099
p=0.2813
p=0.9401
p=0.1411
p=0.4015
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2